

March 06, 2023
STAT490 [Spring 2023]

Instruction:

- Upload both your word and r-files to your blackboard.
 - Make sure to answer clearly all the FIVE questions.
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Meta-analysis of *short term low dose prednisolone versus placebo in rheumatoid arthritis*. The hypothetical table compares **mean grip strength** (mm Hg).

Study	Prednisolone			Placebo		
	Mean	SD	Number	Mean	SD	Number
1	150	55	13	299	55	150
2	213	136	24	249	115	24
3	191	112	20	185	160	20
4	356	151	8	267	267	28
5	109	47	21	97	97	21
6	73.1	43.5	18	59.2	59.2	18

Effect size:

- Mean Difference (MD)

Question:

- 1) [4 points] Analyse the data using the `metacont` function of “`meta`” package
 - a. Use both “`print`” and “`summary`” command to produce your results.
 - b. Produce a forest plot
`metacont (n1, mean1, sd1, n2, mean2, sd2, sm="MD", method.tau="DL")`
- 2) [2 points] Step-by-step produce the following for each study
 - a. ES estimate
 - b. Var(ES)
 - c. 95% CI for your ES
 - d. Use `cbind` to combine your ES, LCL, UCL
- 3) [5 points] COMBINE the studies with **fixed-effect model**
 - a. Calculate the weights
 - b. Total weight (sum of all weights)
 - c. Calculate the relative weight
 - d. Calculate the summary estimate ($M = \text{sum}(\text{rel.weight} * \text{md})$)
 - e. Construct 95% CI for the summary ES M
 - f. Calculate p-value (two-sided)

- 4) [5 points] COMBINE the studies with **Random-effects model**
- a. First calculate tau-sq
 - i. $Q = \sum(w \cdot md^2) - (\sum(w \cdot md))^2 / \text{tot.w}$
 - ii. $C = \text{tot.w} - \sum(w^2) / \text{tot.w}$
 - iii. $Df = k - 1$
 - iv. $\text{tau-sq} = (Q - df) / C$
 - b. Calculate the weight for the random effect
$$wR = 1 / (\text{var.md} + \text{tau}^2);$$
 - c. Total weight (sum of all weights)
$$\text{tot.wR} = \sum(wR);$$
 - d. Calculate the relative weight
$$\text{rel.wR} = wR / \text{tot.wR}$$
 - e. Calculate the summary estimate ($M = \sum(\text{rel.wR} \cdot md)$)
 - f. Construct 95% CI for the summary ES M
 - g. Calculate p-value (two-sided)
- 5) [4 points] Explain, in case if you have seen any difference between results from
- *"meatcont"* and your *step-by-step* analysis
 - *Fixed-effect model* and *random-effects model*.